NETFLOW

LEVEL CONTROL · ROD AND ROPE TYPE

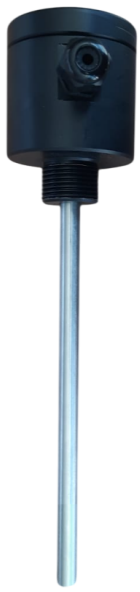
ROD-TYPE DIGITAL CAPACITIVE SENSOR

High-performance level sensor that detects the presence of material in silos, bunkers, hoppers and pipes

This product detects whether product / material is present at the level where it is installed in silos, bunkers, hoppers, pipes, etc. Main target materials: grain, feed, powdered food, plastics, sand, liquids, etc. The output set point is adjusted with the trimpot, or with the display on display models; all parameters are configured using the 2-digit digital display and the MODE / ▲ / ▼ buttons on the board.

12 ... 30 VDC SUPPLY · MAX. 40 mA	**300 mA PNP / Relay** OUTPUT · N.O. / N.C. SELECTABLE
–40 ... +80 °C OPERATING TEMPERATURE · 70% RH	**INSERTION LENGTH** STANDARD 200 / 300 / 400 mm (extendable on request)

- Detection of liquids, powders and flakes
- Adjustable hysteresis and response time
- SS304 stainless steel rod, insulated from housing
- 4-step fine sensitivity adjustment
- Offset calibration for product residue
- Flange screw mounting

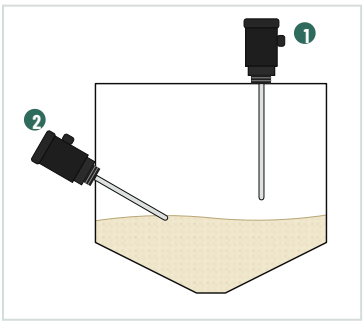


HOUSING VIEW

TECHNICAL SPECIFICATIONS

Measuring principle	Capacitive
Application	Liquids, powders, flakes / products
Supply	12 ... 30 VDC
Current consumption	Max. 40 mA
Output	P-type transistor (PNP) + output, max. 300 mA · N.O. / N.C.
Electrical connection	Cable gland entry, internal 16 – 28 AWG terminal / socket

Display	2-digit digital display, output LEDs
Adjustment	Set trimpot · MODE / ▲ / ▼ buttons
Operating conditions	–40 °C ... +80 °C / max. 70% RH
Housing	Delrin
Rod	SS304 stainless steel rod, insulated from housing
Mounting	Flange, screw-fixed

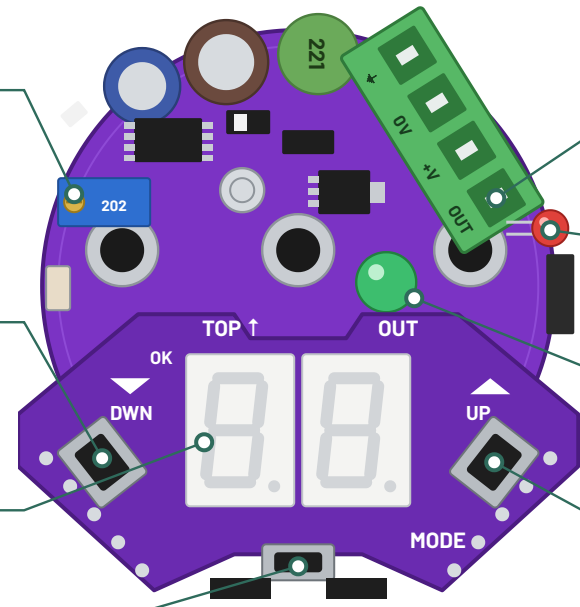


INSTALLATION AND APPLICATION

1. **Vertical mounting.** From the silo roof; the switching point is set by the rod length. Suitable for powder and flake applications.
2. **Downward-angled mounting.** In liquid applications, avoiding contamination at the housing joint improves measurement reliability.

CAUTION For liquid measurement, the power supply earth / PE must be isolated from the system chassis.

INTERNAL VIEW
DISPLAY MODEL · ELECTRONIC BOARD



SET TRIMPOT
Output set point adjustment

▼ DWN BUTTON
Decrease value

2-DIGIT DISPLAY
Measurement, parameter and value

MODE BUTTON
Menu / parameter navigation

TERMINAL
⏚ Earth · 0V · +V · OUT

OUTPUT WARNING LED
Red · output active

OUTPUT STATUS LED
Green · OUT

▲ UP BUTTON
Increase value

TERMINAL CONNECTION

⏚	Earth (GND)
0V	Supply – (12 ... 30 VDC)
+V	Supply + (12 ... 30 VDC)
OUT	PNP + output, max. 300 mA

SETTINGS MENU

01 Press and hold MODE for 4 – 5 s to enter the setting modes.

02 The parameter number and its value are displayed in turn.

03 Change the value with the ▲ / ▼ buttons.

04 Press MODE repeatedly to move to the next parameter.

PARAMETERS

NO	PARAMETER	RANGE	DESCRIPTION
P1	**Set point**	5 – 98	In vertical applications, defines the output set point according to the position of the product in contact with the probe rod surface.
P2	**Threshold / hysteresis**	1 – 20	If required, a +/– threshold / hysteresis value is applied to the set point.
P3	**Fine sensitivity**	1 – 4	Set according to the specific density of the product. Min. sensitivity: liquids and similar · Max. sensitivity: low-density products such as flaked feed.
P4	**Response time filter**	1 – 8	Detection / output delay. 1: no delay · 2: ~1 s · 3: ~2 s · 4: ~3 s · 5: ~4 s · 6: ~5 s · 7: ~6 s · 8: ~7 s
P5	**Output type**	N.O. / N.C. Relay / Transistor	Normally open or normally closed is selected. The output is a P-type transistor + output, max. 300 mA.
P6	**Offset calibration**	21 / 10	Sets the display to "0" when there is an obstacle or product residue build-up.

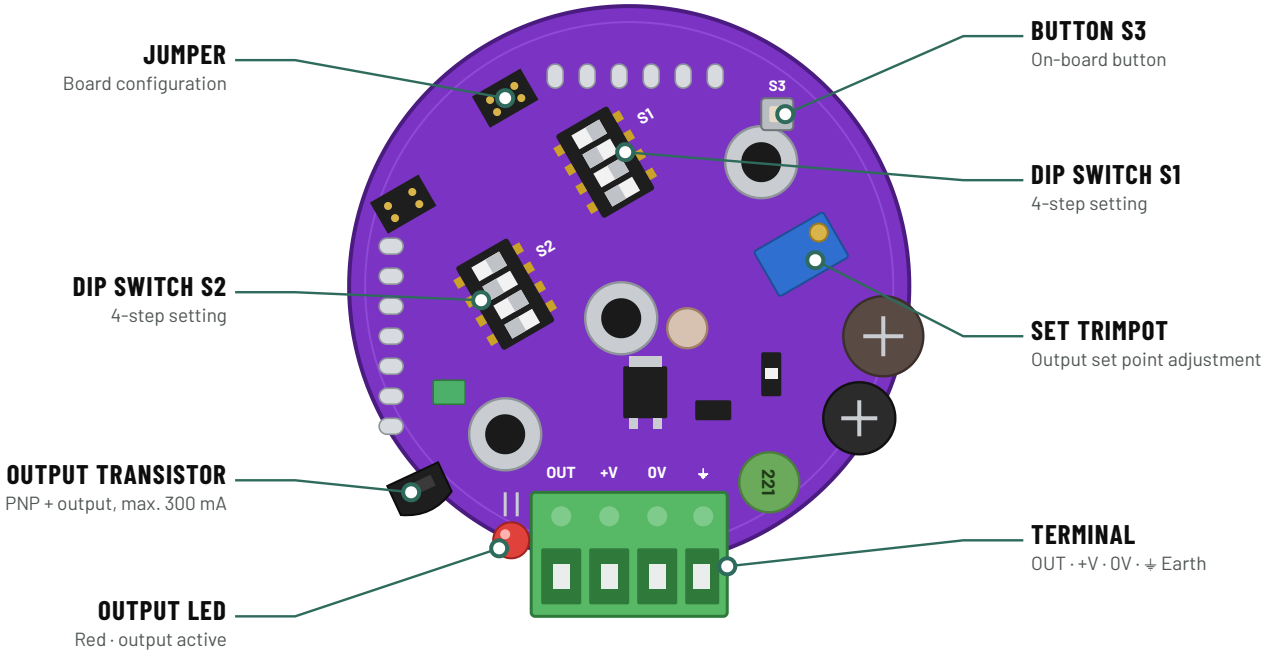
c.o.

CALIBRATION · P6 = 21
Exit with MODE or wait 10 – 15 s for automatic calibration. On success the display shows "c.o." (calibration OK).

c.r.

FACTORY RESET · P6 = 10
Use the same procedure to cancel the calibration. On success the display shows "c.r." (calibration reset).

INTERNAL VIEW
TRANSISTOR MODEL · ELECTRONIC BOARD



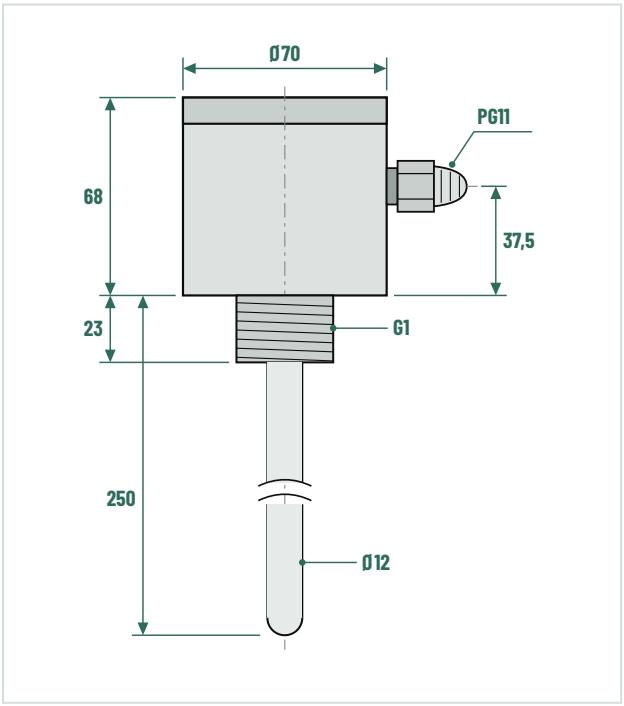
TERMINAL CONNECTION

OUT	PNP + output, max. 300 mA
+V	Supply + (12 ... 30 VDC)
0V	Supply – (12 ... 30 VDC)
⏚	Earth (GND)

ADJUSTMENT ELEMENTS

Set trimpot	Output set point adjustment
S1 · S2	4-way DIP switch
S3	On-board button
LED	Red · output status

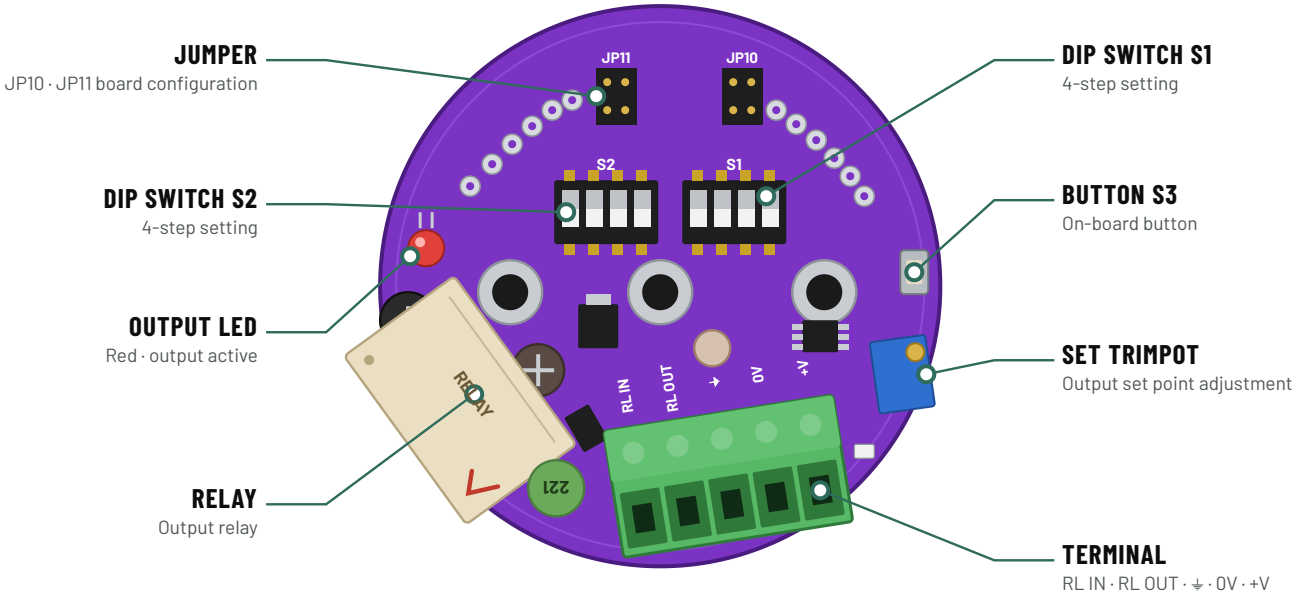
DIMENSIONS
TRANSISTOR MODEL · DIMENSIONS mm



Housing diameter	Ø70 mm
Housing height	68 mm
Cable entry	PG11 · 37.5 mm from housing bottom
Process connection	G1 · 23 mm
Rod	Ø12 mm · 250 mm in drawing

Rod length varies by order; all dimensions in mm.

INTERNAL VIEW
RELAY MODEL · ELECTRONIC BOARD



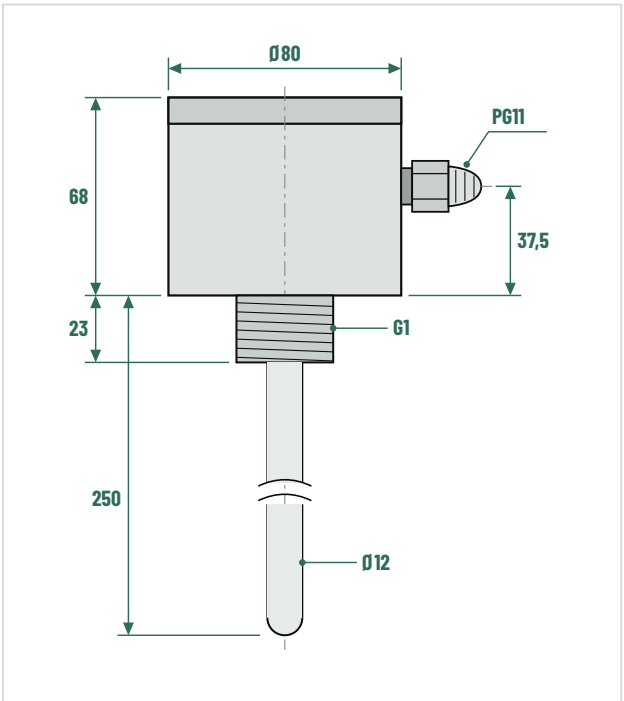
TERMINAL CONNECTION

RL IN	Relay contact terminal
RL OUT	Relay contact terminal
⏚	Earth (GND)
0V	Supply – (12 ... 30 VDC)
+V	Supply + (12 ... 30 VDC)

ADJUSTMENT ELEMENTS

Trimpot	Output set point adjustment
S1 · S2	4-way DIP switch
S3	On-board button
LED	Red · output status
Relay	Contact output (RL IN – RL OUT)

DIMENSIONS
RELAY MODEL · DIMENSIONS mm



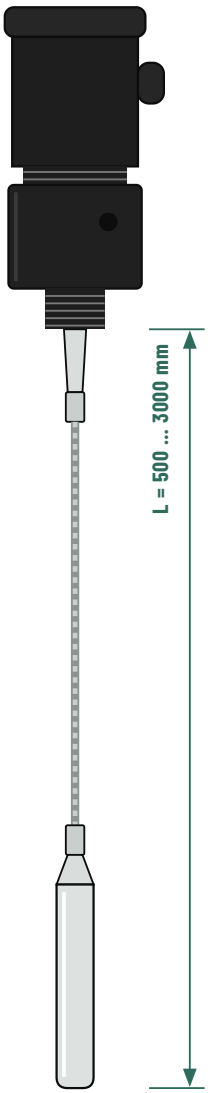
Housing diameter	Ø80 mm
Housing height	68 mm
Cable entry	PG11 · 37.5 mm from housing bottom
Process connection	G1 · 23 mm
Rod	Ø12 mm · 250 mm in drawing

Rod length varies by order; all dimensions in mm.

SUB-MODEL

ROPE-TYPE CAPACITIVE SENSOR

Long-range level detection in tall silos and bunkers



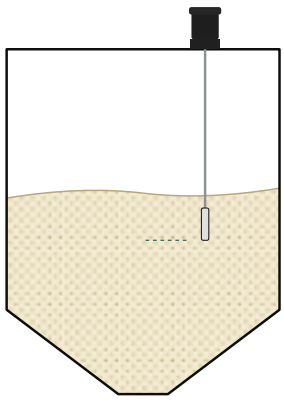
HOUSING VIEW

The rope type uses the same electronic head as the rod-type capacitive sensor; instead of a fixed rod, the sensing element is an end weight attached to a steel rope. The rope hangs vertically from the sensor housing into the silo, and the sensing point is lowered from the silo roof to the required depth. When the product reaches the end weight, the resulting change in capacitance is detected and the output is activated.

500 ... 3000 mm ROPE LENGTH · MIN. / MAX.	**Galvanized steel** ROPE AND END WEIGHT
SS316L / SS304 OPTIONAL STAINLESS STEEL	**12 ... 30 VDC** SUPPLY · SAME AS ROD MODEL

TECHNICAL SPECIFICATIONS

Measuring principle	**Capacitive**
Sensing element	**Steel rope + end weight**
Rope length	**Min. 500 mm · max. 3000 mm**
Rope material	**Galvanized steel · optional SS316L / SS304**
End weight material	**Galvanized steel · optional SS316L / SS304**
Electronics	**Display / transistor / relay model**
Supply · output	**Same as rod model (see page 01)**



INSTALLATION AND APPLICATION

1. **Vertical mounting.** The sensor is mounted on the silo roof; the rope must hang freely and vertically.
2. **Sensing point.** The rope length is selected between 500 – 3000 mm according to the depth at which the level is to be detected.
3. **Position.** The rope and end weight must not be below the filling stream or near areas where product can build up.

CAUTION Water ingress into the sensor housing must be prevented. See pages 02 – 04 for electrical connection and parameter settings.